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Advanced Predictive Modeling of Type 2 Diabetes Using XGBoost and Explainable AI

Zahra Rafie
Yasuj University of Medical Sciences
Moslem Sedaghat Talab
Yasuj University of Medical Sciences
Behrooz Ebrahim Zadeh Koor
Yasuj University of Medical Sciences
Ali Garavand
Lorestan University of Medical Sciences
Cirruse Salehnasab
cirruse.salehnasab@gmail.com

Assistant Professor of Medical Informatics, Yasuj University of Medical Sciences

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Article

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Abstract

The increasing prevalence of Type 2 diabetes (D.M. II) globally poses significant public health challenges, necessitating the development of effective predictive models for accurate prediction. This study aims to apply machine learning (ML) algorithms and explainable artificial intelligence (XAI) techniques to predict the risk of D.M. II using health data from the Dena Cohort in Yasuj, Iran. Data was collected from 3,203 individuals aged 35 to 70, incorporating various demographic, clinical, and lifestyle features. Two ML models, XGBoost and CatBoost, were developed and evaluated for their predictive accuracy. Data preprocessing involved handling missing values, normalization of continuous variables, and addressing class imbalance through the Synthetic Minority Over-sampling Technique (SMOTE). The performance of the models was assessed using accuracy, F1-score, and area under the receiver operating characteristic curve (AUC). The values of SHAP (Shapley Additive Explanations) were utilized to explain model predictions and enhance interpretability. The XGBoost model demonstrated superior performance, achieving an accuracy of 96.07% and an AUC of 99.29%. Key predictive factors identified included fasting blood sugar, fatty liver, urolithiasis, age, and lifestyle factors like energy drink consumption and television watching. The application of SHAP provided valuable insights into the contribution of individual features to the model's predictions, enhancing transparency and understanding for healthcare professionals. This study's findings highlight the potential of ML and XAI in improving the prediction of D.M. II. By identifying critical risk factors, the developed predictive models can support personalized healthcare interventions, improving patient outcomes and reducing healthcare burdens associated with diabetes. This research advocates integrating advanced predictive analytics into clinical practice to enhance diabetes prediction strategies.

Introduction and Literature Review

1.1 Background and Importance of Type 2 Diabetes (D.M. II)

Type 2 diabetes (D.M. II) is a chronic metabolic disorder that represents one of the most significant public health concerns worldwide. Characterized by insulin resistance and progressive beta-cell dysfunction, D.M. II is responsible for considerable morbidity and mortality due to its association with various long-term complications. These complications include cardiovascular diseases, kidney failure, neuropathy, and retinopathy, leading to diminished quality of life and increasing healthcare costs. According to global estimates, the number of people living with diabetes continues to rise rapidly, with projections indicating that over 640 million individuals will be affected by 2040[1–3].

The burden of D.M. II is particularly pronounced in low- and middle-income countries, where healthcare systems often lack the resources for effective prediction and management. The transition towards urbanized lifestyles, marked by reduced physical activity and unhealthy diets, has contributed to increased obesity rates and, subsequently, the prevalence of D.M. II. Moreover, genetic predisposition and environmental factors also play a role in accelerating the onset of the disease. Despite advances in

treatment, many individuals remain undiagnosed until complications arise, emphasizing the need for improved predictive tools that can identify high-risk individuals early [2–5].

1.2 Challenges in Prediction

One critical challenge in managing D.M. II is identifying individuals at risk before the disease progresses to a more advanced stage. Traditional risk assessment tools, though useful, often fail to capture the complex interplay of genetic, environmental, and behavioral variables that influence diabetes risk. This underscores the need for more advanced, data-driven approaches to predict high-risk individuals effectively [6–8].

1.3 The Role of Machine Learning and Artificial Intelligence in Healthcare

In recent years, machine learning (ML) and artificial intelligence (AI) have emerged as promising tools in healthcare, particularly for predictive analytics [9, 10]. ML algorithms excel at processing large datasets, identifying complex patterns, and making accurate predictions that may not be apparent through traditional statistical methods. These capabilities make ML especially valuable for conditions like D.M II, where multiple variables interact to influence disease onset [9–12].

Several ML algorithms, including supervised learning models like decision trees, random forests, and gradient-boosting machines, have been applied to predict D.M. II in various populations. These models are trained on datasets containing input variables (e.g., demographic information, clinical measurements) and the corresponding outcomes (e.g., D.M. II diagnosis). Once trained, these models can predict whether new individuals are at risk of developing D.M. II, enabling proactive identification and intervention [13–15].

1.4 The Need for Explainable AI (XAI)

While ML models can achieve high predictive accuracy, their complexity often renders them "black boxes," making it difficult for healthcare professionals to understand how predictions are made. In clinical settings, trust and transparency are critical. Healthcare providers must understand the reasoning behind an ML model’s predictions to ensure that its recommendations align with clinical judgment and are appropriate for individual patients. This is where explainable AI (XAI) becomes crucial [16–18].

XAI refers to techniques designed to make AI models more transparent and interpretable. Using XAI, the reasoning behind an ML model's predictions can be clearer to clinicians and patients alike. These fosters trust in the model and allows healthcare providers to validate its predictions against their expertise. In the context of D.M. II, XAI can highlight the factors contributing to a patient’s predicted risk, such as elevated blood glucose levels and lifestyle factors, making it invaluable for informed decision-making [17, 19].

1.5 Review of Existing Studies on ML and AI in D.M II Prediction

Several studies have explored the application of artificial intelligence (AI) and machine learning (ML) in the prediction and management of Type 2 diabetes (D.M II). Nomura et al. (2021) conducted a comprehensive review of the role of AI in diabetes management, highlighting that AI has significant potential to enhance diabetes management through predictive analytics [20]. Similarly, Channa et al. (2021) assessed the clinical applications of AI for diabetic retinopathy, demonstrating the efficacy of AI systems in improving screening accuracy [9].

Sheng et al. (2022) provided insights into AI's role in ocular diseases related to diabetes, indicating that AI can effectively assist in predicting and managing ocular complications [11]. Gunasekeran et al. (2020) further explored AI's capabilities in predicting diabetic retinopathy outcomes and found that AI tools significantly improve the prediction of diabetic retinopathy, facilitating timely intervention [1].

Dankwa-Mullan et al. (2019) investigated how AI can transform diabetes care delivery through a systematic review of AI applications. They concluded that AI technologies can improve access, accuracy, and efficiency in diabetes care [2]. Contreras and Vehi (2018) reviewed AI methods for decision support in diabetes care, emphasizing that AI can enhance clinical decision-support systems, ultimately improving diabetes management outcomes [5].

Bhat et al. (2023) developed a framework for predicting diabetes risk using ML techniques, finding that their comprehensive approach effectively predicts diabetes risk [21]. Ahmed et al. (2022) investigated the effectiveness of fused ML techniques in predicting diabetes and noted improved prediction accuracy by integrating diverse machine-learning approaches [22].

Khanam and Foo (2021) compared various ML algorithms for diabetes prediction and highlighted the effectiveness of ensemble methods [23]. Finally, Sisodia et al. (2018) evaluated the effectiveness of classification algorithms for diabetes prediction, concluding that certain algorithms provide high accuracy in predicting D.M. II [24].

1.6 Gaps in Research and Future Directions

Despite the significant advances in ML and XAI for D.M II prediction, areas still require further exploration. Most existing studies focus on a narrow set of risk factors, whereas future research could benefit from incorporating more comprehensive data, such as social determinants of health and lifestyle data. Additionally, while XAI techniques have improved model transparency, more work is needed to make these explanations actionable for healthcare providers. Bridging the gap between AI predictions and clinical decision-making remains a challenge.

As AI continues to evolve, future research should aim to integrate ML models into clinical practice in a way that complements, rather than replaces, the expertise of healthcare professionals. By developing

more interpretable and accessible models, AI can become a valuable tool for preventing D.M. II and improving patient outcomes.

Materials and Methods

2.1 Study Design

This research is a cross-sectional study aimed at predicting Type 2 diabetes (D.M II) using machine learning (ML) algorithms and explainable artificial intelligence (XAI) techniques. Data was gathered from the Dena Cohort in Yasuj, Iran, consisting of 3,203 individuals aged 35 to 70. The study followed six stages: identification of variables and risk factors, data collection, data preprocessing, model development, evaluation of machine learning models, and interpretability using XAI methods.

2.2 Study Population

The data for this study was collected from the Dena Cohort in Yasuj, Iran, between March 2020 and November 2022. The cohort comprises health data from individuals who provided information on demographics, medical history, clinical measurements, and lifestyle factors. In this study, individuals with incomplete or missing critical data were excluded. A total of 3,203 individuals were included in the analysis.

2.3 Variables and Risk Factors

The study considered several demographic, clinical, and lifestyle variables for predicting D.M. II. Some of the most relevant variables identified were:

- **Demographic variables**: Age, gender, employment status.
- **Clinical variables**: Fasting blood sugar (FBS), triglycerides, liver and kidney health (e.g., fatty liver, urolithiasis), and red blood cell count.
- **Lifestyle factors**: Consumption of energy drinks, fruit juice, coffee, tea, and screen time (e.g., watching television).

Feature selection was carried out using the LightGBM algorithm to rank the importance of each feature in predicting D.M. II. Redundant or less important features were eliminated to streamline the models.

2.4 Data Preprocessing

Several preprocessing steps were implemented to ensure the data quality and accuracy of the models:

- **Handling missing data**: Cases with significant missing data were excluded, and for minor missing entries, imputation techniques were used.

- **Normalization**: The min-max scaling technique normalized continuous variables like FBS and triglycerides.
- **Outlier detection and removal**: Outliers were identified using interquartile range (IQR) analysis and were removed to reduce noise in the data.
- **Balancing the dataset**: The class imbalance between diabetic and non-diabetic individuals was addressed using the Synthetic Minority Over-sampling Technique (SMOTE), ensuring the model was not biased toward the majority class [25].
- **Feature Selection**: In the final preprocessing stage, the LGBM (Light Gradient Boosting Machine) algorithm was employed for feature selection. This method is known for its efficiency and effectiveness in identifying the most relevant features from the dataset, which can significantly enhance the performance of machine learning models [26]. Utilizing LGBM for feature selection, we aimed to reduce dimensionality, improve model accuracy, and streamline the training process.

2.5 Machine Learning Models

Two machine learning models were used for prediction:

- **XGBoost (XGBClassifier)**: A highly efficient gradient-boosting algorithm known for its performance with structured data. It builds models iteratively, optimizing the loss function at each step [10].
- **CatBoost (CatBoostClassifier)**: An algorithm particularly well-suited for categorical features. It prevents overfitting and increases accuracy through ordered boosting and various data pre-processing techniques [27].

Both models were trained using 80% of the data, while 20% was reserved for testing. Ten-fold cross-validation was employed to prevent overfitting.

2.6 Model Evaluation and Hyperparameter Tuning

The models were evaluated using the following metrics:

Accuracy

The proportion of correct predictions (true positives and negatives) out of all predictions. It is calculated as

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Where:

- TP (True Positives) refers to the cases where the model correctly predicted D.M II (positive) cases.
- TN (True Negatives) refers to the cases where the model correctly predicted non-D.M. II (negative) cases.
- FP (False Positives) refers to non-D.M. II cases incorrectly classified as diabetic.

- FN (False Negatives) refers to D.M. II cases incorrectly classified as non-diabetic.

F1-Score

The harmonic means of precision and recall, balancing the trade-off between false positives and false negatives. It is especially useful for imbalanced datasets. The formula for F1-score is

$$\mathrm{F1-Score} = 2^{*}\frac{Precision * \ Recall}{Precision + \ Recall}$$

Precision and recall are defined as:

$$Precision = \frac{TP}{TP + FP}$$

$$Recall = \frac{TP}{TP + FN}$$

AUC-ROC (Area Under the Receiver Operating Characteristic Curve)

This metric measures the model's ability to distinguish between positive (D.M II) and negative (non-D.M. II) cases. A higher AUC value indicates better model performance.

To optimize the models' performance, a random search technique was employed for hyperparameter tuning. The key hyperparameters adjusted included the learning rate, maximum depth, and the number of estimators.

2.7 Explainable AI (XAI) Techniques

The SHAP (Shapley Additive Explanations) method enhanced the predictive models' interpretability. SHAP values explain the contribution of each feature to a model's prediction by assigning importance scores, both at the global and individual levels. This approach enhances transparency, allowing healthcare professionals to understand the factors driving each prediction [28].

2.8 Software Tools

The study employed several software tools for model development and evaluation:

- **Python 3.13**: Used for implementing machine learning models and preprocessing.
- **Pycaret**: A key library for building machine learning pipelines.
- **XGBoost and CatBoost**: Libraries used for gradient boosting implementations.
- **SHAP**: Used for interpreting the output of the machine learning models.

2.9 Ethical Considerations

The study was conducted under ethical standards for research involving human participants. Informed consent was obtained from all participants in the Dena Cohort, and all data were anonymized before analysis. The ethics review board of the relevant institution granted ethical approval.

Results

3.1 Data Preprocessing

Data from 3,203 individuals aged between 35 and 70 was extracted from the Dena Cohort database. Of these, 402 individuals (12.55%) were diagnosed with Type 2 Diabetes (D.M II), while 2,801 individuals (87.45%) were non-diabetic. Several preprocessing steps were performed on this dataset. Initially, variables with less than 95% variance were removed, and records with more than 50% missing values were excluded. Missing data was then imputed using the "most frequent" method. After these steps, the final dataset consisted of 3,203 records with 80 features (Table 1).

Table 1
Features and specifications of the Dataset

Row	Feature	Description	Type	Role
1	GenderID	Male or Female	Independent	Input
2	Age	Numeric	Independent	Input
3	LastEduID	Highest Educational Degree	Independent	Input
4	MET_Final	Metabolic condition of individuals	Independent	Input
5	WaistCircumference	Waist circumference (numeric)	Independent	Input
6	HipCircumference	Hip circumference (numeric)	Independent	Input
7	WristCircumference	Wrist circumference (numeric)	Independent	Input
8	BMI	Body mass index based on height and weight	Independent	Input
9	HasJob_new	Recent job change	Independent	Input
10	Employmentstatus	Employment status	Independent	Input
11	SleepDuration24h1	Sleep duration in 24 hours	Independent	Input
12	SleepDurationMidDay1	Midday nap duration	Independent	Input
13	TV1	Hours of TV watching	Independent	Input
14	AtDeskWork1	Desk job hours	Independent	Input
15	Computer1	Computer usage hours	Independent	Input
16	Eating1	Amount of food consumed	Independent	Input
17	Cooking1	Method of food preparation	Categorical	Input
18	Driving1	Time spent driving	Independent	Input
19	Walking1	Time spent walking	Independent	Input
20	AerobicExercise1	Time spent on aerobic exercise	Independent	Input
21	DrivingHeavyVehicle1	Heavy vehicle driver	Independent	Input
22	LightAgricultural1	Low-maintenance agricultural work	Independent	Input
23	HeavyLaborAgricultJobs1	High-maintenance agricultural work	Independent	Input
24	DPB	Diastolic blood pressure	Independent	Input
25	SPB	Systolic blood pressure	Independent	Input
26	RBC	Red blood cell count	Independent	Input

Row	Feature	Description	Type	Role
27	HGB	Hemoglobin level	Independent	Input
28	FBS	Fasting blood sugar	Independent	Input
29	TG	Triglycerides level	Independent	Input
30	CHOL	Cholesterol level	Independent	Input
31	SGOT	Aspartate aminotransferase (AST) enzyme	Independent	Input
32	SGPT	Alanine aminotransferase (ALT) enzyme	Independent	Input
33	ALP	Alkaline phosphatase enzyme	Independent	Input
34	HDL.C	High-density lipoprotein cholesterol (HDL-C)	Independent	Input
35	GGT	Blood glucose after glucose intake and specific time	Independent	Input
36	LDL_Calc	Low-density lipoprotein cholesterol (LDL-C)	Independent	Input
37	WSI_total	Number of imaging sessions	Independent	Input
38	UseDrugs	Use of narcotics or opiates	Independent	Input
39	UseAlcohol	Alcohol consumption	Independent	Input
40	Pizza	Pizza consumption	Independent	Input
41	TEACOFFEE	Tea and coffee consumption	Independent	Input
42	BAVARAGE	Energy drinks consumption	Independent	Input
43	Pickle	Pickle consumption	Independent	Input
44	Oil	Oil consumption	Independent	Input
45	SimpleSugar	Simple sugar consumption	Independent	Input
46	Vegetables	Vegetable consumption	Independent	Input
47	Nuts	Nut consumption	Independent	Input
48	Juice	Juice consumption	Independent	Input
49	Fruits	Fruit consumption	Independent	Input
50	Legumes	Legume consumption	Independent	Input
51	Whitemeat	White meat consumption	Independent	Input

Row	Feature	Description	Type	Role
52	Redmeat	Red meat consumption	Independent	Input
53	Totalmeat	Total meat consumption	Independent	Input
54	dairyproduct	Dairy product consumption	Independent	Input
55	Wholegrain	Whole grain consumption	Independent	Input
56	Nutrient FFQ_protein	Daily protein intake	Independent	Input
57	Total_lipid_fat	Liquid fat consumption	Independent	Input
58	GrilledFoodIntID	Grilled food consumption	Independent	Input
59	FriedFoodIntID	Fried food consumption	Independent	Input
60	PotatoFryTypeID	Fried potato consumption	Independent	Input
61	VegFryTypeID	Fried vegetable consumption	Independent	Input
62	UsedOilTypeID	Type of oil used	Independent	Input
63	ReUseMold	Reuse of containers	Independent	Input
64	UsedScratchedTeflon	Use of scratched Teflon pans	Independent	Input
65	FoodSaltUsedID	Salt used in food	Independent	Input
66	OnionFryTypeID	Fried onion consumption	Independent	Input
67	CookingWareID1	Type of cookware used	Independent	Input
68	VegKeepDried	Consumption of dried vegetables	Independent	Input
69	VegKeepRefrigerator	Consumption of refrigerated vegetables	Independent	Input
70	VegKeepFreezer	Consumption of frozen vegetables	Independent	Input
71	BreadContainerTypeID1	Bread-containing products consumption	Independent	Input
72	LemonContainerTypeID1	Lemon-containing products consumption	Independent	Input
73	CVD_History	History of stroke	Independent	Input
74	HasDepression	Presence of depression	Independent	Input
75	urolithiasis	Presence of urolithiasis	Independent	Input
76	HasThyroidDisease	Presence of thyroid disease	Independent	Input
77	HasFattyLiver	Presence of fatty liver	Independent	Input

Row	Feature	Description	Type	Role
78	HasCardiacDisease	Presence of heart disease	Independent	Input
79	HasHypertension	Presence of hypertension	Independent	Input
80	Has D.M II	Presence of Type 2 diabetes	Independent	Target

Due to the class imbalance between diabetic and non-diabetic cases, the SMOTE (Synthetic Minority Over-sampling Technique) method was applied to balance the dataset. This increased the sample size to 5,602 records, with 2,801 cases of diabetes and 2,801 non-diabetic cases. Among the 80 features, 79 were independent variables (Table 1), and one was the dependent variable (D.M II diagnosis). The dataset was then randomly split into an 80% training set (4,481 records) and a 20% test set (1,121 records).

3.2 Performance of Machine Learning Models

Two machine learning models, XGBoost and CatBoost, were developed to predict the likelihood of D.M. II. The models' performance was evaluated using three key metrics: accuracy, F1-score, and AUC (Table 2 and Fig. 1).

- **XGBoost Model**: The XGBoost algorithm yielded the highest predictive performance, with an accuracy of 96.07%, an F1-score of 96.10%, and an AUC of 99.29%.
- **CatBoost Model**: The CatBoost classifier demonstrated slightly lower performance, with an accuracy of 95.31%, an F1-score of 95.31%, and an AUC of 99.01%.

Table 2
Evaluation results of machine learning models' performance

No.	Machine Learning Models	Accuracy (%)	F1-Score (%)	AUC (%)	Average (%)
1	XGBClassifier	96.07	96.1	99.29	97.1
2	CatBoostClassifier	95.31	95.31	99.01	96.54

Overall, the XGBoost model was identified as the best-performing model based on its superior accuracy and AUC. This model was further evaluated using a confusion matrix, class prediction error, and feature importance to ensure robustness and interpretability (Fig. 2).

3.3 Key Predictive Factors

The study identified ten key predictive factors for D.M. II. The most significant factors were:

1. **Fasting Blood Sugar (FBS)**: Elevated FBS levels, particularly within the prediabetes range (100 to 126 mg/dL), were the strongest predictor of D.M. II, accounting for 41% of the model’s predictive power.
2. **Fatty Liver**: Individuals with fatty livers contributed 21% to the prediction of D.M. II.

3. **urolithiasis**: It was identified as a significant factor, contributing 15%.
4. **Age**: Advancing age accounted for 4% of the predictive ability.
5. **Consumption of Energy Drinks**: This lifestyle factor contributed 2.5%.
6. **Fruit Juice Consumption**: The intake of various fruit juices contributed 2.4%.
7. **Triglycerides (TG)**: Elevated triglyceride levels contributed 2.3%.
8. **Watching Television**: Extended periods of TV viewing accounted for 2.3%.
9. **Coffee and Tea Consumption**: The intake of these beverages contributed 2.2%.
10. **Red Blood Cell (RBC) Count**: An increase in RBC count contributed 2% to the risk of developing D.M. II.

These ten features were identified as the most impactful in predicting D.M. II, with FBS having the highest importance. Figure 3 visualizes a detailed analysis of feature importance.

3.4 Model Interpretability Using SHAP

The SHAP (Shapley Additive Explanations) method enhanced the predictive models' interpretability. SHAP values were used to explain the contribution of each feature to the model’s predictions, providing global and individual insights. For instance, SHAP analysis for individual cases demonstrated how certain factors, such as elevated FBS or consumption of sugary beverages, influenced the likelihood of D.M. II development. This transparency improved trust in the model and provided actionable insights for healthcare professionals (Fig. 4).

Figure 4 presents three cases labeled "Case 1," "Case 2," and "Case 3," each displaying interpretability plots generated by a machine learning model to explain its predictions. These plots use SHAP (SHapley Additive exPlanations) to show how different features contribute to the model's decision.

Case 1

The top plot highlights that " HasFattyLiver" (with a positive contribution of 1.544) and " FBS" (0.5672) increase the model's prediction, while "Age" (-1.5) and other features pull it downward.

Case 2

The middle plot shows that "urolithiasis" (1.324) has a positive influence, whereas " FBS" (-0.6688) and other variables like " BAVARAGE" significantly affect the prediction in either direction.

Case 3

The bottom plot illustrates the influence of " Juice" (0.3342) increasing the prediction, while features such as " FBS" and "TEA/COFFEE" contribute negatively.

The visualizations demonstrate various features' positive and negative impacts on the model's predictions in each case. The machine learning model produced accurate predictions for D.M. II, with specific features influencing individual risk.

Discussion and Conclusion

4.1 Implications of Machine Learning in Type 2 Diabetes Prediction and Healthcare Transformation

This study demonstrates the effectiveness of machine learning (ML) models, particularly XGBoost, in predicting Type 2 diabetes (D.M. II) with remarkable accuracy. With an accuracy of 96.07% and an AUC of 99.29%, XGBoost outperformed traditional models by analyzing large datasets and capturing complex interactions among numerous predictive variables. Fasting blood sugar (FBS) was identified as the strongest predictor, aligning with established research that associates elevated FBS with diabetes risk. Other significant predictors, such as fatty liver and urolithiasis, underscore the impact of metabolic comorbidities on D.M. II development, while lifestyle factors like energy drink consumption and prolonged television watching highlight the contribution of lifestyle choices to disease risk [1, 2, 5, 9, 11, 20–24, 29, 30].

The study also identified a connection between red blood cell (RBC) parameters and D.M. II, revealing that lower RBC counts and elevated red cell distribution width (RDW) are associated with poor glycemic control. This finding aligns with prior research suggesting that RBC health may be related to diabetes complications. Including RBC metrics in predictive models may improve the accuracy of diabetes risk assessments and provide a more comprehensive understanding of patients' health profiles. This broader scope of risk factors allows for a more holistic prediction model that integrates metabolic, lifestyle, and physiological dimensions of diabetes risk [31–34].

The application of SHAP (Shapley Additive Explanations) values in this study enhanced the interpretability of the ML models, which is crucial for clinical adoption. SHAP allows clinicians to understand and explain the contribution of individual risk factors in the model's predictions, fostering greater trust in AI-driven decision-making. This transparency is invaluable in patient-provider communication, allowing healthcare providers to offer clear explanations for recommended lifestyle or treatment interventions. Furthermore, SHAP’s interpretability gives ML models an advantage over traditional statistical approaches, enabling more nuanced, patient-specific recommendations tailored to individual health data.

A significant implication of this study is its support for a shift from reactive to proactive, preventative healthcare. By integrating predictive models into routine clinical workflows, healthcare systems can identify high-risk individuals earlier, implementing timely and targeted interventions to prevent D.M. II onset and complications. This transition could ultimately reduce the healthcare and economic burdens associated with diabetes by prioritizing early, data-informed management strategies that lessen the need for intensive interventions later.

Beyond D.M. II, this research paves the way for predictive models to be applied in managing various chronic diseases, which share risk profiles related to metabolic and lifestyle factors. Expanding ML-based predictive analytics in healthcare can enhance resource allocation, streamline patient care, and ensure that at-risk populations receive early, personalized attention. The scalability of these models means they could be adapted to a range of health conditions, thereby promoting more equitable and efficient healthcare delivery across diverse populations.

4.2 Limitations and Future Directions

While promising, the study's findings are limited by reliance on data from the Dena Cohort, which may impact the model's generalizability across different populations. Additionally, the use of SMOTE to address class imbalance introduces synthetic data that may not fully represent real-world cases. Future research should validate these models across diverse populations, explore alternative balancing methods, and incorporate longitudinal data from wearable devices to support real-time, dynamic risk assessments. Expanding datasets to include genetic and metabolic biomarkers could further enhance model accuracy and applicability in personalized medicine. Additionally, incorporating real-time health data from wearable devices, such as continuous glucose monitors and activity trackers, would allow for adaptive tools that provide timely, personalized insights into diabetes risk.

4.3 Conclusion

This study illustrates the potential of ML models, combined with explainable AI (XAI), to transform D.M. II prediction and management by identifying key risk factors and offering clear, interpretable explanations for each prediction. By integrating such predictive tools into clinical practice, healthcare providers can adopt a more proactive, data-informed approach to patient care, ultimately improving outcomes and reducing the long-term burden of diabetes on healthcare systems. Although limitations exist, this study provides a robust foundation for future advancements in predictive modeling and AI-driven diabetes care. As AI continues to evolve, its integration into clinical workflows holds substantial promise for enhancing the prediction, prevention, and management of chronic diseases like D.M. II, offering a transformative impact on healthcare.

Declarations

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Author's Contribution

Zahra Rafie: Visualization, Investigation, Resources, Writing - Original Draft. **Moslem Sedaghat Talab**: Conceptualization, Visualization, Validation, Investigation, Writing - Review & Editing. **Behrooz Ebrahim**

Zadeh Koor: Investigation, Data Curation, Writing - Review & Editing. **Ali Garavand**: Methodology, Software, Formal analysis, Writing - Review & Editing. **Cirruse Salehnasab**: Conceptualization, Project administration, Supervision, Funding acquisition, Validation, Formal analysis, Writing - Review & Editing

Conflicts of Interest

The authors declare no conflicts of interest regarding the publication of this study.

Financial Disclosure

No financial interests related to the material of this manuscript have been declared.

Data availability statement

The datasets analyzed during the current study are available from the corresponding author on reasonable request. The Python code of this project is available via this link:

https://github.com/salehnasab/Advanced-Predictive-Modeling-of-Type-2-Diabetes/blob/main/DMPrediction.ipynb

Ethics Approval

The Iran National Committee approved this study for Ethics in Biomedical Research with Approval ID IR.YUMS.REC.1402.152.

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Figures

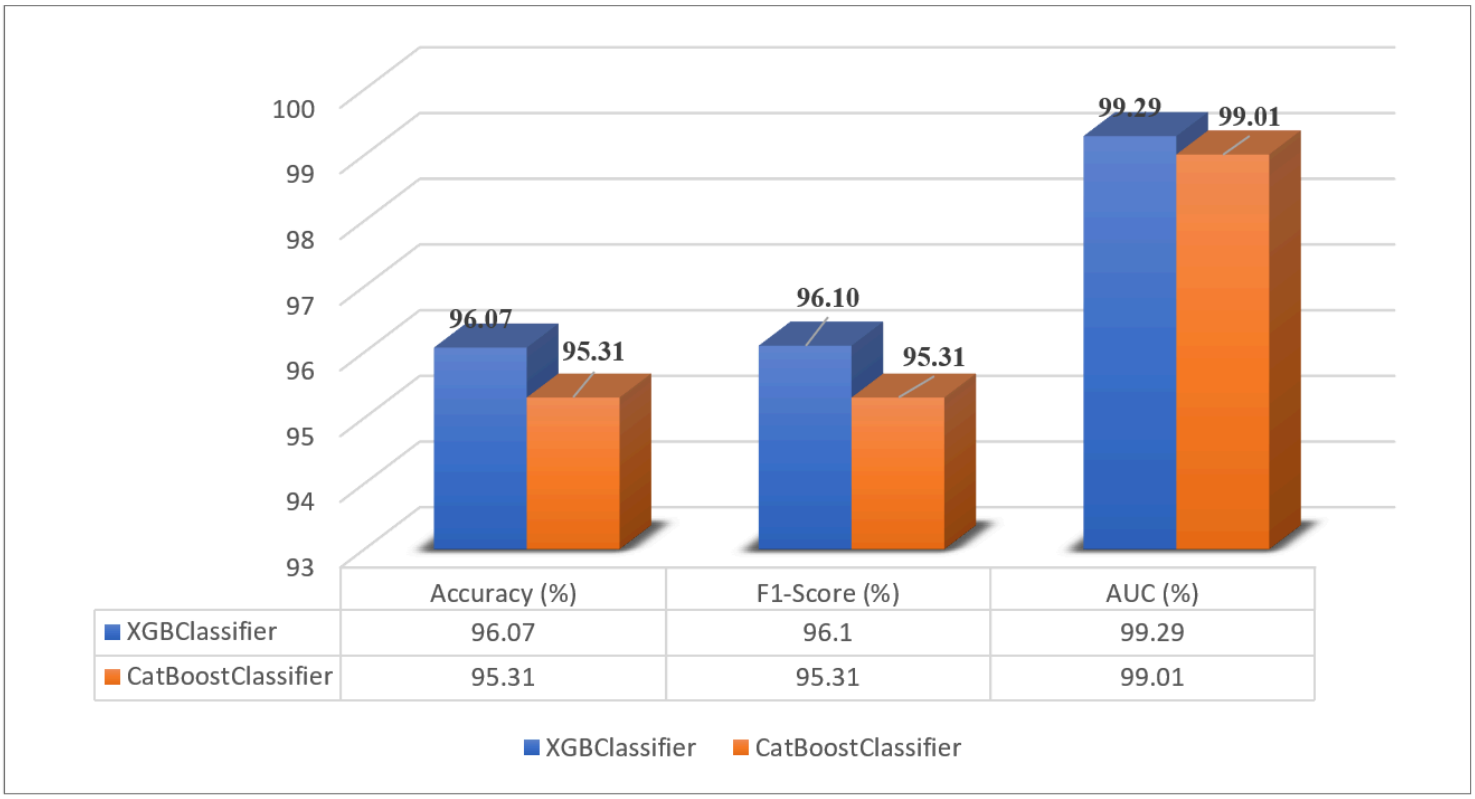

Figure 1

Evaluation results of machine learning models' performance

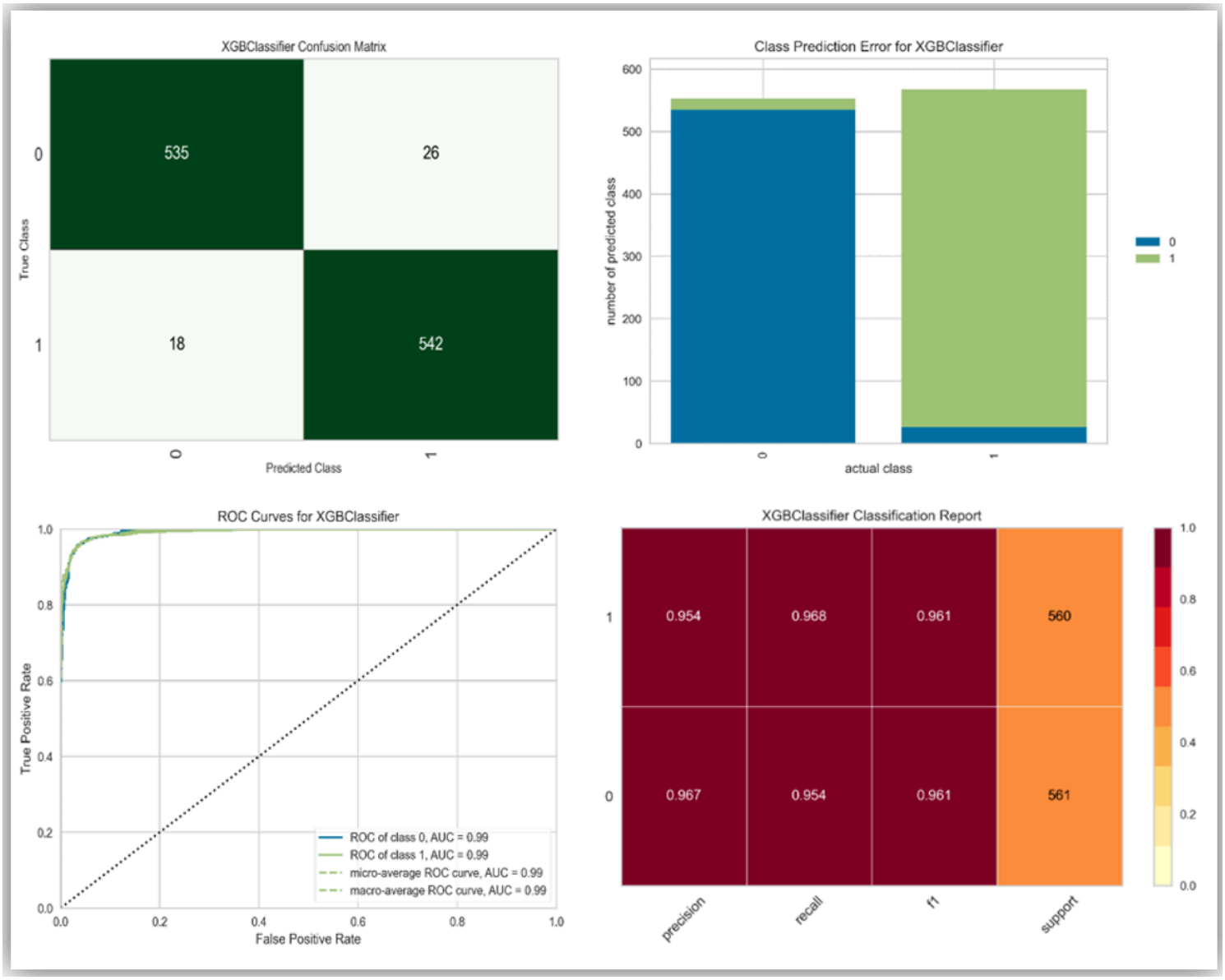

Figure 2

Comprehensive Performance Evaluation of the XGBClassifier Model

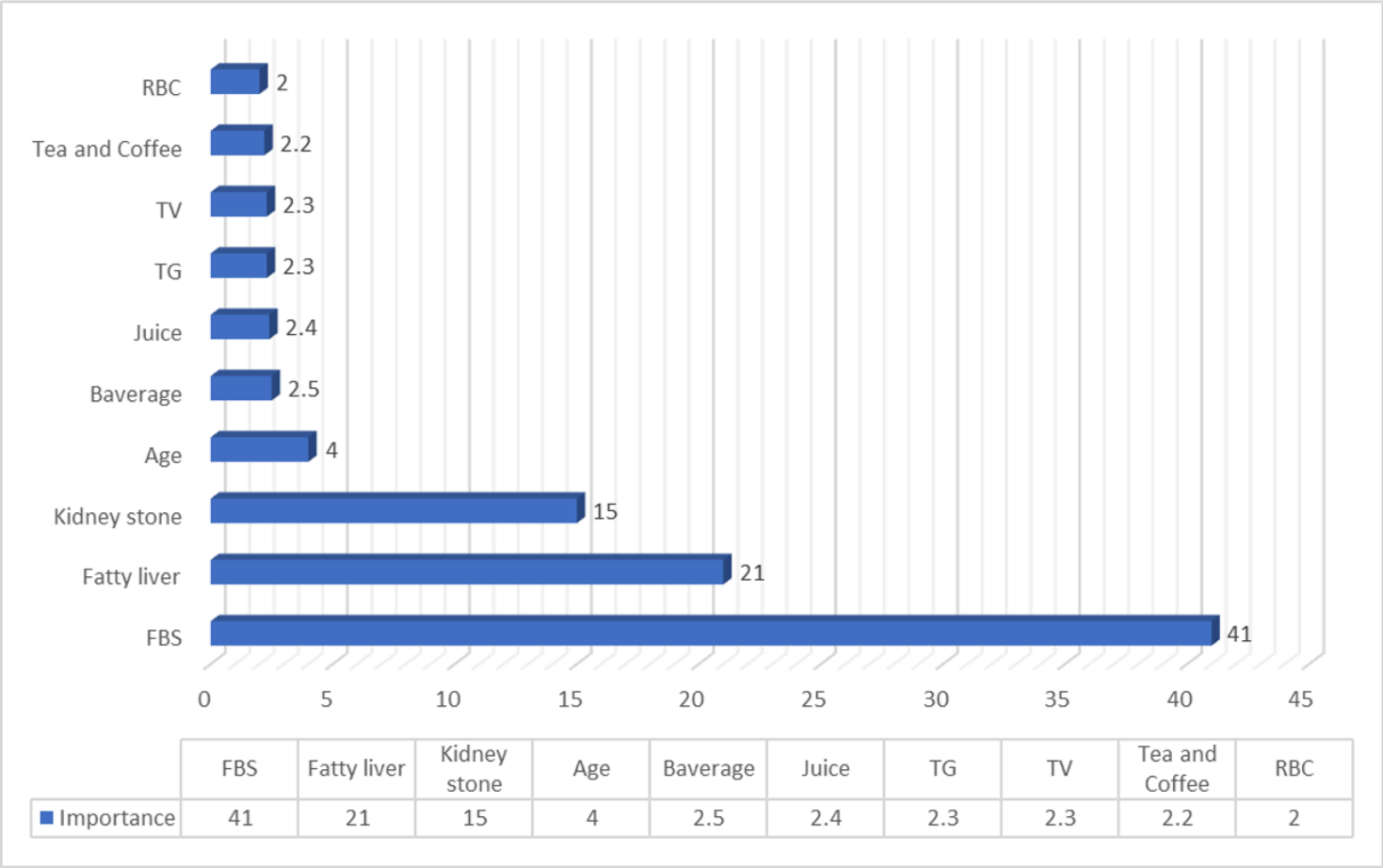

Figure 3

Key Predictors for Assessing the Risk of Type 2 Diabetes Using the LGBM Algorithm

Figure 4

The SHAP values that are associated with the input data for three cases